

Charles “Trey” Werr

Software Engineer

Contact

- (913) 749-2176
- treywerr@gmail.com
- <https://github.com/treywerr>

Education

- University of Kansas
2018 - 2022
Honors Program
Bachelor's Degree
Computer Science

Languages

- C#
- C++
- HLSL
- Java

Hard Skills

- Unity Engine
- Object-Oriented Programming
- Computer Graphics Programming
- Debugging
- Profiling and Optimization
- Research & Development
- Algorithms
- Mathematics
- Physics

Soft Skills

- Project Management
- Stakeholder Communication
- Requirements Elicitation
- Cross-discipline Collaboration

Summary

Software Engineer with 4+ years of experience in Unity and C# development. Experienced in research and development, systems architecture design, and performance optimization. Specialized in algorithms for real-time manipulation of 3D models. Seeking a long term position within a stable engineering organization building interactive or simulation-driven software.

Experience

Level 1 Researcher | Team Lead

Acadental Inc. | September 2022 - Present

- R&D lead for technical systems on a VR training simulation program deployed on Meta Quest
- Communicated with stakeholders to implement features to desired specifications
- Coordinated team members on active work and managed release cycle
- Simulated realistic tissue deformation using position-based dynamics
- Created a radiograph X-ray rendering system based on raytracing
- Developed a rigidbody collision / depenetration system for VR interaction
- Employed shape registration and voxelization algorithms to compare 3D scan models
- Engaged in methodical profiling and optimization to meet strict performance requirements
- Implemented a flexible JSON-based data serialization system used in networking, saves, and recording